

# Zuo Gou

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## EDUCATION

### Tsinghua University

Sep 2023 – Expected Jun 2028

B.Eng., Automation (Xinya College & Dept. of Automation)

Beijing, China

- **GPA: 3.8/4.0**    **Awards:** Tsinghua Comprehensive Scholarship (2024, 2025); Luo Yuehua Scholarship (2024)

## MANUSCRIPTS

- [1] **Z. Gou** et al. *Relative Noise Schedules in Joint Continuous–Discrete Diffusion*. In preparation; target ICML 2027.
- [2] **Z. Gou** et al., W.-C. Ma. *Humanoid2Humanoid: Shared Sensing for Cooperative Loco-Manipulation*. In preparation; target CVPR 2027.

## RESEARCH EXPERIENCE

### Prof. Wei-Chiu Ma's Group (Dept. of Computer Science, Cornell University)

Jun 2026 – Nov 2026

Research Intern, Multi-Agent Humanoid Locomotion & Inter-Robot Communication

Ithaca, NY

- Studying how two humanoids can share what they see in real time, so that a robot walking backward climbs stairs hidden from its own cameras by relying on its partner's view.
- Established when such sharing matters: robots usually infer hidden terrain from gait and contact, and only in specific conditions does the partner's view become indispensable, recovering 20–40 success points.
- Built decentralized policies that act on their own sensing plus messages from the partner, halt safely when the link drops and resume when it returns; demonstrated a simulated pair climbing stairs together.

### Prof. Ge Liu's Group (Dept. of Computer Science, University of Illinois Urbana-Champaign)

Mar 2026 – Present

Research Intern, Diffusion Models & Generative Modeling Theory

Remote

- Studying how a diffusion model that generates continuous and discrete quantities at once — atom positions with atom types and bonds — should pace noise across its two channels, a choice now hand-tuned per dataset.
- Characterized when the likelihood bound is indifferent to that pacing and what the best schedule looks like when it is not, and confirmed the predictions on public molecular models.
- Showed that reported likelihoods are dominated by a single coarse step of the sampler, and built an estimator reaching the same precision with roughly  $2,600\times$  less computation.

### Prof. Huazhe Xu's Group (IIS, Tsinghua University)

Jul 2025 – Jan 2026

Research Assistant, Vision–Language–Action Models & RL

Beijing, China

- With Chenhao Lu, co-developed *PlugRL* ([plugrl.github.io](https://github.com/plugrl); open source, ongoing), a wire protocol between environments and RL trainers that lets environments on other machines, in other languages, or outside a trainer's cluster still be trained; wrote the protocol specification and its conformance checkers.

### Prof. Yebin Liu's Group (Dept. of Automation, Tsinghua University)

Sep 2024 – Jun 2025

Research Assistant, Dexterous Manipulation & Diffusion-based Control

Beijing, China

- Developed a diffusion model that generates physically plausible hand–object interaction for dexterous manipulation, with the simulation environments and demonstration pipelines it was trained on.

## TECHNICAL SKILLS

### Programming

Python (PyTorch), C++, CUDA, MATLAB, Shell

### Robotics & simulation

IsaacGym, Isaac Lab, MuJoCo, Legged Gym, RSL-RL

### Infrastructure

Linux, Slurm clusters, distributed multi-GPU training, Weights & Biases